

A Reproducible Open-Source Framework for Comparative MPPT Evaluation Across Perturbative, Rule-Based, Learning-Based, and Model-Based Control Paradigms Under Partial Shading

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Abstract—Maximum power point tracking (MPPT) comparison studies are a saturated literature: most either compare perturb-and-observe (P&O), incremental conductance (IncCond), and fuzzy logic variants, or propose yet another bio-inspired metaheuristic optimizer. This paper instead compares five MPPT algorithms spanning four genuinely distinct design paradigms – perturbative (P&O, IncCond), rule-based (fuzzy logic), learning-based (tabular Q-learning), and model-based nonlinear control (sliding mode control, SMC, with a Lyapunov stability guarantee) – under a common, open-source, pure-Python simulation framework. The framework uses a two-diode photovoltaic model with bypass-diode-induced partial shading, validated against a commercial Canadian Solar CS6P-250P datasheet, driving a boost converter under 11 test conditions (7 single-module scenarios plus 4 partial-shading configurations), each evaluated over 50 Monte Carlo runs with identical random seeds across algorithms. Results show that sliding mode control achieves the highest steady-state tracking efficiency (99.90%) and lowest oscillation, but all five algorithms – including Q-learning – fail to converge within 2% of the true global maximum power point in the majority of complex partial-shading patterns tested, with perturbative and model-based algorithms converging instead to a shared local maximum and Q-learning (never trained on multi-module shading) failing in a qualitatively different, worse way. A reduced 5×5 fuzzy rule base is shown to statistically match the full 7×7 base’s tracking efficiency, a rule-reduction result with direct computational relevance given fuzzy logic’s measured $\sim 101 \times$ per-step computational cost relative to P&O. All code, data, and figures are released under an open-source license for full reproducibility.

Index Terms—Maximum power point tracking, photovoltaic systems, sliding mode control, fuzzy logic control, reinforcement learning, partial shading, Monte Carlo simulation, reproducible research.

I. INTRODUCTION

Maximum power point tracking (MPPT) is essential to extracting the available power from a photovoltaic (PV) module or array, whose I-V characteristic has a single global maximum under uniform illumination but can develop multiple local maxima under partial shading once bypass diodes begin conducting. Decades of MPPT research have produced a wide

design space of tracking strategies; the seminal comparison in this same journal, [1], surveyed 19 such techniques, but as a narrative survey rather than a paradigm-structured, statistically-validated comparison, and predates both learning-based and model-based nonlinear control as MPPT approaches entirely. In the two decades since, the comparative literature built around that design space has instead converged on two recurring, increasingly saturated patterns: (i) re-comparing perturb-and-observe (P&O), incremental conductance (IncCond), and fuzzy logic control against one another, and (ii) proposing yet another bio-inspired metaheuristic optimizer (grey wolf, particle swarm, whale, dung beetle, and so on), a pattern reviewers have grown visibly fatigued by. Neither pattern, on its own, demonstrates genuine coverage of the MPPT design space, and few published comparisons release open, reproducible code that a reviewer or reader can independently re-run.

This paper takes a different approach on both axes. First, rather than adding a fifth variation on perturbative search or a new metaheuristic, we compare five algorithms chosen specifically to span four *qualitatively distinct* MPPT design paradigms:

- **Perturbative** – P&O with adaptive step size, and IncCond, representing the classical baseline and its refinement;
- **Rule-based** – a fuzzy logic controller (FLC) with a physically-justified 49-rule (7×7) base and centroid defuzzification;
- **Learning-based (model-free)** – tabular Q-learning, trained on a subset of irradiance conditions and evaluated separately on held-out conditions to directly test generalization rather than memorization;
- **Model-based nonlinear control** – sliding mode control (SMC) with a sliding surface on $dP/dV = 0$, an exponential reaching law, boundary-layer chattering mitigation, and an explicit Lyapunov stability argument.

Second, every result in this paper is produced by a fully open-source, pure-Python simulation framework – no MATLAB/Simulink dependency – with pinned dependencies, a continuous-integration test suite, and Monte Carlo statistical

rigor (50 independent runs per algorithm-scenario pair, using identical random seeds across algorithms for a fair paired comparison). The underlying photovoltaic model is a two-diode model with bypass-diode composition, quantitatively validated against a commercial Canadian Solar CS6P-250P datasheet rather than an idealized or unvalidated model, and the partial-shading test scenarios are constructed to produce two and three local maxima, directly probing each algorithm's global-search capability rather than only its local convergence speed.

A. Contributions

This paper makes the following contributions:

- 1) A cross-paradigm comparison of five MPPT algorithms spanning perturbative, rule-based, learning-based, and model-based control, rather than variations on a single paradigm;
- 2) An open-source, pure-Python, CI-tested simulation framework (two-diode PV model with bypass diodes, boost converter with state-space small-signal model, unified algorithm interface, Monte Carlo scenario runner) released for full reproducibility;
- 3) Explicit, separately-reported train-distribution and held-out evaluation for the learning-based algorithm, directly addressing the generalization-versus-memorization question that pooled reporting would obscure;
- 4) A quantitative finding that all five algorithms – across all four paradigms, including the model-based SMC controller with its Lyapunov-guaranteed local stability – fail to reach the true global maximum power point in the majority of the complex partial-shading patterns tested, with the specific failure mode differing by paradigm (Section VII);
- 5) A fuzzy rule-base sensitivity analysis showing a reduced 5×5 rule base statistically matches the full 7×7 base's tracking efficiency, a result with direct computational-cost relevance given fuzzy logic's substantially higher per-step cost relative to the other four algorithms.

The remainder of this paper is organized as follows. Section II positions this work against prior MPPT comparison literature, organized by the same four paradigms used above. Section III and Section IV describe the photovoltaic and boost-converter models and their validation. Section V details all five algorithm implementations. Section VI defines the test scenarios, Monte Carlo methodology, and metrics. Section VII presents statistical results and figures. Section VIII concludes with limitations and future work.

II. LITERATURE REVIEW

This review is organized by the same four MPPT design paradigms compared in this paper, both to survey prior work and to make the cross-paradigm framing of Section I concrete, followed by recent cross-paradigm reviews that position this paper's own contribution.

A. Perturbative Methods

Perturb-and-observe remains the most widely deployed MPPT method for its simplicity and low computational cost; [2] provides the adaptive-step-size design this paper's P&O implementation follows, trading off oscillation amplitude against tracking speed by scaling the perturbation step with $|dP/dV|$. Incremental conductance [3] refines this by directly comparing dI/dV against $-I/V$, in principle eliminating steady-state oscillation around the true maximum power point (MPP) at the cost of a boundary-condition singularity as $dV \rightarrow 0$ that must be handled explicitly. Adaptive step-size tuning for both methods remains an active research area: [4] uses particle swarm optimization to tune the step size of both P&O and IncCond for a grid-tied system, reporting improved steady-state oscillation over fixed-step baselines – the same tradeoff this paper's adaptive-step P&O design targets, though via a simpler analytical rule rather than a metaheuristic-tuned one.

B. Rule-Based Methods

Fuzzy logic control removes the need for an explicit analytical model of the PV/converter system by encoding tracking behavior as a linguistic rule base. [5] surveys four FLC-MPPT input/output variable choices ($dP&dV$, $dP&dI$, $E&dE$, $G&F$) at a 5×5 partition size; this paper's design instead uses a 7×7 partition following common $(E, \Delta E) \rightarrow \Delta D$ conventions in the wider fuzzy-MPPT literature [6]. [7] presents a hybrid fuzzy logic MPPT controller specifically evaluated under partial shading, reporting improved global-search behavior over a standalone fuzzy controller – directly relevant context for this paper's own finding (Section VII) that a non-hybrid fuzzy controller, like every other non-hybrid algorithm tested, fails to escape a local maximum under partial shading; a hybrid extension is exactly the kind of change that finding motivates as future work.

C. Learning-Based Methods

[8] is the foundational reference for model-free reinforcement learning applied to MPPT, using tabular Q-learning over a discretized (voltage, dP/dV) state space – the same state representation this paper's Q-learning implementation uses. [9] studies Q-table design specifically, proposing an improved update scheme for the same tabular setting this paper uses rather than moving to function approximation or deep RL – a natural point of comparison for whether this paper's Q-learning generalization failure (Section VII-B) is a consequence of the particular Q-table design or a more fundamental limitation of tabular discretization under distribution shift. A key open question for any learning-based MPPT controller is whether it generalizes to conditions outside its training distribution or merely memorizes them; [10] studies a deep-RL controller specifically under partial shading, a harder and more directly relevant generalization test than flat-irradiance held-out evaluation alone, and is a natural point of comparison for this paper's own partial-shading generalization finding (Section VII), where the Q-learning agent – trained only on

flat, single-module irradiance – fails on multi-module shading in a qualitatively different way than the perturbative and model-based algorithms.

D. Model-Based Nonlinear Control

Sliding mode control (SMC) applies variable-structure control theory [11] to MPPT by defining a sliding surface (commonly $s = dP/dV$ [12], driving the system toward the MPP condition) and a reaching law governing convergence to that surface. This paper’s exponential reaching law follows [13], and its boundary-layer chattering mitigation – replacing $\text{sign}(s)$ with a saturation function – follows the classical treatment in [14]. Chattering mitigation remains an active SMC-MPPT research area: [15] combines PID-type SMC with an improved grey wolf optimizer to tune controller gains rather than relying on a hand-tuned boundary layer as this paper does, illustrating one direction (automated gain tuning) this paper’s grid-searched `SMC_K/SMC_Q/SMC_BOUNDARY` constants (`config.py`) could be extended toward. Per CLAUDE.md’s own caution, no single PV-SMC paper combines this exact surface, reaching law, and boundary-layer choice at once (a search for one during this literature review turned up several partial matches – e.g. a multi-power reaching law with a sigmoid switching function – but none matching all three design choices simultaneously); the design lineage is instead cited piece-by-piece: $s = dP/dV$ as the common surface choice [12], the exponential reaching law from [13], and the boundary-layer chattering mitigation from [14].

E. PV Modeling and Boost Converter Design

The boost converter sizing and small-signal analysis in this paper follow the standard treatment in [16], including the right-half-plane zero in the control-to-output transfer function that simplified worked examples sometimes omit. [17] addresses boost-converter control specifically in discontinuous conduction mode (DCM) under a constant-power load – directly relevant to this paper’s own DCM finding (Section IV), that this design’s converter enters DCM at light load and that every algorithm’s duty-cycle-perturbation reasoning implicitly assumes the CCM small-signal model; [17]’s dedicated DCM control strategy is one concrete way a future revision could remove that confound rather than merely flag it as a limitation.

[18] and [19] both address PV array modeling but at different levels: the former’s contribution is a single-diode, three-datasheet-point parameter extraction method, while the two-diode model used in this paper – which better represents recombination currents at low voltage and is important for correctly resolving multiple local maxima under partial shading – follows [19] specifically; the two should not be conflated when citing the two-diode model’s origin. More recent parameter-extraction work continues to refine this: [20] proposes an analytical (non-iterative) two-diode extraction method, and [21] compares two datasheet-based double-diode extraction approaches on an accuracy/ simplicity tradeoff – directly relevant to this paper’s own extraction method (Section III-C), which is explicitly flagged as a simplification (fixed a_1 , analytically-solved $I_{s1} = I_{s2}$) relative to the full

iterative procedure in [19]; both recent papers are natural points of comparison for tightening that simplification in future work. [22] provides a recent analytical treatment of bypass- and blocking-diode effects on the partial-shading P - V curve, complementing this paper’s simulation-based bypass-diode composition (Section III) with an independent analytical derivation of the same multiple-local-maxima phenomenon this paper’s Monte Carlo results depend on.

F. Cross-Paradigm Positioning

The most direct historical point of comparison is [1], whose 19-technique survey remains, two decades later, the most widely-cited broad MPPT comparison in this journal. Three things distinguish this paper from it, beyond the two decades of algorithm development in between: [1] is a narrative survey of results reported by different authors under different conditions, not a common statistically-validated experimental comparison; it predates learning-based and model-based nonlinear control as MPPT paradigms entirely, so a four-paradigm framing was not yet possible; and it does not release runnable code, so its comparisons cannot be independently re-executed the way every result in this paper can be.

[23] and [24] are both recent, broad reviews spanning classical, metaheuristic, and AI-based MPPT methods – the same cross-paradigm scope this paper argues is missing from most single-comparison studies (Section I). Neither, however, provides the specific combination this paper does: a validated physical model, open-source reproducible code, and Monte Carlo statistical rigor applied uniformly across all paradigms compared, rather than a literature synthesis of results reported under differing conditions by different authors. [25] and [26] are both recent examples of the “yet another bio-inspired/hybrid metaheuristic optimizer” pattern this paper’s introduction explicitly declines to add to (a zebra optimization algorithm and a GWO-PSO hybrid, respectively) – included here not as directly comparable baselines but as evidence the pattern remains active and crowded, reinforcing rather than undermining this paper’s choice to compete on reproducibility and paradigm coverage instead. [27] is a recent, directly relevant example of the open-source reproducibility movement this paper is also part of, releasing a Python framework for metaheuristic MPPT techniques evaluated under partial shading – but, consistent with the pattern above, it is scoped to a single paradigm (metaheuristic optimization) rather than the four-paradigm comparison this paper provides.

III. PHOTOVOLTAIC SYSTEM MODELING

A. Two-Diode Model

The reference module is a Canadian Solar CS6P-250P, a 60-cell polycrystalline module rated at $P_{max} = 250$ W ($V_{oc} = 37.2$ V, $I_{sc} = 8.87$ A, $V_{mp} = 30.1$ V, $I_{mp} = 8.30$ A at standard test conditions, STC: 1000 W/m², 25°C), chosen as a real, commercially available panel rather than an idealized one so simulated results can be validated quantitatively against the manufacturer’s own datasheet.

Each series-connected cell string is modeled by the two-diode equivalent circuit

$$I = I_{ph} - I_{s1} \left[e^{\frac{V+IR_s}{a_1 V_t}} - 1 \right] - I_{s2} \left[e^{\frac{V+IR_s}{a_2 V_t}} - 1 \right] - \frac{V + IR_s}{R_{sh}}, \quad (1)$$

where $V_t = a N_s kT/q$ is the string-level thermal voltage, N_s the number of series cells in the string, k Boltzmann's constant, T the cell temperature in Kelvin, and q the electron charge. Unlike the single-diode model, the second diode term (with ideality a_2 and saturation current I_{s2}) explicitly represents recombination current in the depletion region, which becomes significant at low forward bias – the regime relevant to correctly resolving secondary local maxima under partial shading. We follow [19] for the two-diode formulation; [18]'s widely-cited three-point extraction method is a single-diode technique and is cited here only for general parameter-extraction background, not as the source of (1).

B. Bypass Diodes and Partial Shading

Following standard industry practice, one bypass diode protects every 20 cells, giving $N_s/20 = 3$ bypass groups for the 60-cell reference module. Each group is modeled as an independent two-diode string of (1); when a forced string current would exceed a group's own short-circuit current at its (possibly shaded) irradiance, that group's bypass diode is modeled as an ideal diode with $V_f = 0.7$ V, clamping the group's contributed voltage to $-V_f$ rather than driving it into deep reverse bias. This bypass mechanism is what allows multiple local maxima to appear in a shaded string's P - V curve at all – without it, a series-connected string's power curve remains unimodal regardless of shading, which would make the partial-shading scenarios in Section VI unable to test any algorithm's global-search capability.

C. Parameter Extraction

The four free parameters (I_{ph} , R_s , R_{sh} , a_2) are fit to the datasheet's four STC key points (V_{oc} , I_{sc} , V_{mp} , I_{mp}) via bounded nonlinear least squares (Trust Region Reflective), minimizing the residual of three conditions: the short-circuit current match, the current-at- V_{mp} match, and $dP/dV = 0$ at the reported MPP. The first diode's ideality factor a_1 is fixed at 1.0 (the midpoint of its physically expected range) rather than fit, and $I_{s1} = I_{s2}$ is solved analytically from the open-circuit-voltage boundary condition at every optimizer iteration rather than treated as a free parameter. This leaves the extraction under-determined by one degree of freedom relative to the full iterative procedure in [19]; we flag this explicitly as a simplification rather than presenting it as the complete published procedure.

For the CS6P-250P, this extraction yields $I_{ph} = 8.880$ A, $I_{s1} = I_{s2} = 2.891 \times 10^{-10}$ A, $R_s = 0.308 \Omega$, $R_{sh} = 269.8 \Omega$, $a_1 = 1.000$, $a_2 = 1.406$.

D. Model Validation

Table I compares the simulated I-V curve at the four STC key points against the datasheet. The maximum relative error across all four points is 0.068% (at P_{max}), well within the 2% acceptance criterion.

TABLE I
SIMULATED VS. DATASHEET STC KEY POINTS (CS6P-250P)

Quantity	Simulated	Datasheet	Rel. Error
I_{sc} (A)	8.870	8.870	0.000%
V_{oc} (V)	37.200	37.200	0.000%
V_{mp} (V)	30.100	30.100	0.000%
I_{mp} (A)	8.300	8.300	0.000%
P_{max} (W)	249.83	250.00	0.068%

E. Temperature Dependence

Photocurrent and both diodes' saturation currents are temperature-dependent, following [28]:

$$I_{ph}(G, T) = I_{ph,ref} \frac{G}{G_{ref}} (1 + K_i(T - T_{ref})), \quad (2)$$

$$I_s(T) = I_{s,ref} \left(\frac{T}{T_{ref}} \right)^{3/a} \exp \left[\frac{E_g q}{ak} \left(\frac{1}{T_{ref}} - \frac{1}{T} \right) \right], \quad (3)$$

applied independently to I_{s1} (using a_1) and I_{s2} (using a_2), with $E_g = 1.121$ eV the silicon bandgap energy and $K_i = +0.065\%/^{\circ}\text{C}$ the datasheet's short-circuit-current temperature coefficient. Both correction factors in (2)-(3) are identically 1 at $T = T_{ref}$, so STC behavior is unaffected by this extension – Table I above is unchanged whether or not temperature dependence is included.

This addresses a real error found during development: an earlier version of this model let temperature affect only the thermal-voltage term $V_t(T)$, leaving I_{ph} and I_s temperature-independent. Holding I_s fixed while V_t rises with temperature makes $V_{oc} \approx V_t \ln(I_{ph}/I_s)$ increase with temperature – backwards from every real PV panel, where I_s 's much stronger (exponential) growth with temperature is the physically dominant effect and pulls V_{oc} down. Concretely, the uncorrected model predicted $V_{oc}(50^{\circ}\text{C}) = 40.32$ V against this panel's datasheet-implied 34.04 V (using its linear $K_v = -0.34\%/^{\circ}\text{C}$ coefficient) – the wrong sign, not merely an uncalibrated magnitude. (2)-(3) fix this: the corrected model gives $V_{oc}(50^{\circ}\text{C}) = 34.31$ V, within 0.8% of the datasheet-implied value, and $I_{sc}(50^{\circ}\text{C})$ matches to within 0.01% (the *temperature step* scenario in Section VII is the only one of this paper's eleven scenarios that varies temperature – every other scenario runs at a constant 25°C and is completely unaffected by this fix).

Table II validates the corrected model against the manufacturer's NOCT electrical table ($G = 800$ W/m², $T_c = 45 \pm 2^{\circ}\text{C}$) – data not available until the full datasheet PDF was sourced during this project. RMSE across the five NOCT key points is 1.35%, within the 2% acceptance criterion (the individual P_{max} point alone is 2.44%, narrowly over 2% on its own, though the aggregate RMSE the criterion specifies is not). Low-irradiance validation is necessarily coarser: the datasheet gives only a single retention claim ("±95.5% module efficiency from 1000 to 200 W/m²"), not a full I-V table at 200 W/m², so this is a single ratio check rather than a point-by-point match like NOCT. The model measures 93.5% retention – close to, but about 2 percentage points below, the datasheet's claim,

plausibly because this simplified model holds R_s and R_{sh} irradiance-independent rather than scaling R_{sh} with irradiance as a full five-parameter model would.

TABLE II
NOCT VALIDATION ($G = 800 \text{ W/m}^2$, $T_c = 45^\circ\text{C}$)

Quantity	Simulated	Datasheet	Rel. Error
I_{sc} (A)	7.188	7.190	0.024%
V_{oc} (V)	34.514	34.200	0.917%
V_{mp} (V)	27.814	27.500	1.143%
I_{mp} (A)	6.666	6.600	0.999%
P_{max} (W)	185.41	181.00	2.435%

RMSE across all five points: 1.35%.

IV. BOOST CONVERTER MODELING

A. Specifications and Component Sizing

The MPPT controllers command the duty cycle of a boost converter interposed between the PV module/string and a resistive load representing a downstream DC bus, switching at $f_s = 20 \text{ kHz}$ with an input range $V_{in} \in [20, 40] \text{ V}$ (covering the module's expected V_{mp} range under varying conditions) and output $V_{out} = 48 \text{ V}$. Sizing follows the standard CCM boost relations

$$D = 1 - \frac{V_{in}}{V_{out}}, \quad (4)$$

$$L = \frac{V_{in} D}{f_s \Delta I_L}, \quad \Delta I_L = 0.30 I_{L,avg}, \quad (5)$$

$$C = \frac{I_{out} D}{f_s \Delta V_o}, \quad \Delta V_o = 0.01 V_{out}, \quad (6)$$

with the 30% inductor-current-ripple and 1% output-voltage-ripple design targets following [16]. At the worst-case operating point ($V_{in} = 20 \text{ V}$, full load $P_{max} = 250 \text{ W}$), this gives $D = 0.583$, $L = 155.6 \mu\text{H}$, $C = 316.5 \mu\text{F}$, and load resistance $R = V_{out}^2 / P_{max} = 9.22 \Omega$.

B. Small-Signal Model

The state-space averaged control-to-output transfer function for a CCM boost converter is

$$G_{vd}(s) = \frac{V_{out}}{(1-D)^2} \cdot \frac{1 - s/\omega_z}{1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0}\right)^2}, \quad (7)$$

with

$$\omega_0 = \frac{1-D}{\sqrt{LC}}, \quad Q = (1-D)R\sqrt{\frac{C}{L}}, \quad \omega_z = \frac{(1-D)^2 R}{L}. \quad (8)$$

At the design operating point, $\omega_0 = 1878 \text{ rad/s}$ (299 Hz), $Q = 5.48$, and the right-half-plane (RHP) zero $\omega_z = 10,286 \text{ rad/s}$ (1637 Hz). The RHP zero is included explicitly in (7) because a boost converter's control-to-output response has non-minimum-phase behavior (an initial dip in output before the commanded rise) that a simplified transfer function omitting the zero would misrepresent – of direct relevance to any MPPT algorithm that assumes a monotonic response to a duty-cycle perturbation.

C. Discontinuous Conduction Mode (DCM)

The critical inductance for continuous conduction at a given operating point is $L_{crit} = (1-D)^2 DR / (2f_s)$. At full load this design's $L = 155.6 \mu\text{H}$ exceeds $L_{crit} = 23.3 \mu\text{H}$, confirming CCM operation; at light load (5% of P_{max} , representative of low-irradiance conditions), the same inductor value no longer satisfies the CCM condition and the converter enters DCM.

This has a concrete implication for MPPT behavior that is not simply a converter-design footnote: in DCM, the V_{out}/V_{in} relationship depends on L , f_s , and load in addition to duty cycle, so (7) no longer holds and the same duty-cycle perturbation produces a different, load-dependent voltage/power response than in CCM. Every algorithm in Section V infers tracking direction from a duty-cycle perturbation's effect on dP/dV (P&O, IncCond, fuzzy logic, and SMC's sliding surface directly; Q-learning's reward indirectly) without re-deriving or re-tuning against a DCM model. A spurious sensitivity change at light load is therefore a plausible confound for any algorithm whose convergence or oscillation metrics degrade specifically in this paper's low-irradiance scenarios (200 W/m^2 in the multi-level-irradiance and rapid-double-step scenarios, or the low-irradiance modules in the partial-shading scenarios) – a possibility we return to in Section VIII's discussion of limitations, not something addressable by changing the MPPT algorithm itself.

V. MPPT ALGORITHMS

Every algorithm implements the same interface, $D_{k+1} = f(V_k, I_k, D_k)$, so the scenario runner (Section VI) drives all five without any algorithm-specific special-casing. Duty cycle is clamped to $D \in [0.05, 0.90]$ throughout, keeping the operating point away from the $D \rightarrow 0/D \rightarrow 1$ singularities of the boost converter's $R_{in} = R(1-D)^2$ relation.

A. Perturb and Observe (Adaptive Step)

P&O [2] perturbs duty cycle and observes the resulting power change. Direction is inferred self-referentially – by comparing this step's power change against the sign of the *previous* duty-cycle perturbation – so the algorithm never needs to assume which way voltage moves for a given duty-cycle direction. Step size adapts between a large step (0.01) when $|dP/dV|$ exceeds a threshold (2.0 W/V) and a small step (0.001) near the MPP, trading convergence speed against steady-state oscillation amplitude, per [2]'s adaptive-step design.

B. Incremental Conductance

IncCond [3] compares incremental conductance dI/dV against $-I/V$; at the true MPP these are equal since $dP/dV = I + V dI/dV = 0$. When $|dV|$ is too small to trust that ratio (the boundary case), the sign of dI alone determines direction instead. A fixed duty-cycle step (0.001) is applied once direction is determined.

C. Fuzzy Logic Control

The FLC uses inputs $E = dP/dV$ (normalized to $[-1, 1]$ by an empirically grid-searched gain, since this panel's dP/dV is steeply asymmetric – approximately $+8.9 \text{ W/V}$ near $V = 0$ versus a steep negative rolloff toward -76 W/V approaching V_{oc}) and its step-to-step change ΔE , producing a duty-cycle step ΔD . Membership functions are triangular with saturating shoulder sets at both ends of each universe, and the $7 \times 7 = 49$ -rule base uses seven linguistic labels (NB, NM, NS, Z, PS, PM, PB) per input.

Rather than hand-transcribing 49 rules, the rule base is *generated* from a formula of our own design (not adapted from a specific prior publication), chosen to reproduce a conventional error/change-of-error $\rightarrow$ duty-step rule table entry-for-entry while remaining parameterizable to the 5- and 3-label reduced bases used later in this section:

$$D_{\text{level}} = \text{clip} \left(\text{round} \left(-\frac{E_{\text{level}} + \Delta E_{\text{level}}}{2} \right) \right), \quad (9)$$

where levels are signed integers counted from the center label ($Z = 0$) to the extremes (NB/PB = ∓ 3). Equation (9) states the physical intent common to every rule: the further the operating point is from the MPP and the faster it is moving away, the larger and more oppositely-signed the corrective duty-cycle step should be. Applying the same formula to 5- and 3-label sets produces internally-consistent reduced rule bases used for the sensitivity analysis in Section VII. Defuzzification uses the centroid method over a 201-point discretization of the output universe.

D. Q-Learning

Following [8], the state space is the discretized pair $(V, dP/dV)$: voltage into 30 bins over $[0, V_{oc,STC}]$, and dP/dV into 7 bins after normalization by the same empirically-grounded scale used for the fuzzy controller's E input. This discretization is coarse by deliberate choice, not oversight: it keeps tabular Q -learning tractable to train (Section VIII returns to whether finer discretization or function approximation would narrow the held-out generalization gap reported in Section VII, a question this paper's fair-comparison design does not attempt to answer with the same tabular architecture). The action space is $\{-\delta, 0, +\delta\}$ on duty cycle ($\delta = 0.005$), mirroring P&O's action set for a fair cross-paradigm comparison. The reward is $r_k = P_k - P_{k-1}$ (change in power, not raw power): using raw power produced a policy that got permanently stuck several volts past the true MPP for some training irradiances, because coarse voltage-bin discretization can keep the operating point in the same state bin for several consecutive steps while it drifts away from the MPP, and a reward tied only to that bin's power level does not register the drift; reward-on- ΔP penalizes power-losing actions immediately regardless of which bin they land in. Training uses standard Q -learning with ϵ -greedy exploration (decaying from $\epsilon = 1.0$ to $\epsilon = 0.05$), learning rate $\alpha = 0.15$, and discount $\gamma = 0.9$, cycling round-robin over training irradiances $\{600, 800, 1000\} \text{ W/m}^2$ at fixed STC temperature and a single, uniformly-illuminated module – i.e.,

training never exercises multi-module partial shading or non-STC temperature at all. *The methodological consequence of this is central to Section VII:* every scenario is explicitly tagged as train-distribution or held-out relative to this training regime (Section VI), and results for the two are never pooled.

E. Sliding Mode Control

SMC uses the sliding surface $s = dP/dV$ (zero exactly at the MPP, needing no explicit reference voltage) and the exponential reaching law [13]

$$\dot{s} = -k \text{sign}(s) - q s, \quad k, q > 0, \quad (10)$$

combining a constant-rate term (drives $s \rightarrow 0$ in finite time, dominant far from the surface) with a proportional term that vanishes at $s = 0$, avoiding the overshoot a pure constant-rate law alone would cause. The discontinuous $\text{sign}(s)$ term is the classic source of chattering; it is replaced by a boundary-layer saturation $\text{sat}(s/\Phi, \beta)$ [14] – linear (and therefore continuous) for $|s/\Phi| \leq \beta$ and clipped to ± 1 outside it. Mapping (10) onto a discrete duty-cycle correction, and accounting for the boost converter's $R_{in} = R(1-D)^2$ (increasing duty cycle decreases operating voltage, so $s > 0$ – operating point left of the MPP, needing higher V – requires *lower* D), gives

$$\Delta D = -(K \text{sat}(e, \beta) + Q e), \quad e = \text{clip}(s/\Phi, -1, 1). \quad (11)$$

a) *Lyapunov stability.*: Taking $\mathcal{V}(s) = \frac{1}{2}s^2$ as the Lyapunov candidate (positive definite, zero only at $s = 0$), along the continuous-time reaching law (10),

$$\dot{\mathcal{V}} = s \dot{s} = s(-k \text{sign}(s) - qs) = -k|s| - qs^2 \leq 0, \quad (12)$$

with equality only at $s = 0$. Since $k, q > 0$, $\mathcal{V}$ strictly decreases whenever $s \neq 0$, so $s \rightarrow 0$ asymptotically: the sliding surface – and therefore the MPP condition $dP/dV = 0$ – is reached and maintained. Within the boundary layer, replacing $\text{sign}(s)$ with the saturation function trades this strict guarantee for practical, chattering-free convergence to a small residual band around $s = 0$, the standard boundary-layer tradeoff [14]. This paper's implementation empirically confirms the mitigation is load-bearing, not merely present: shrinking the boundary layer toward pure $\text{sign}(s)$ reproduces the classic chattering problem (0% steady-state power ripple at the design boundary width versus sustained $\sim 2.5\%$ ripple at a boundary width two orders of magnitude narrower).

VI. TEST SCENARIOS, MONTE CARLO METHODOLOGY, AND METRICS

A. Test Scenarios

Eleven test scenarios are used, seven single-module conditions plus four partial-shading configurations (three complex patterns count as separate scenarios since each represents a distinct shading pattern, not a parameter sweep of one scenario): *steady-state* (1000 W/m^2 , 25°C , 2 s), *multi-level irradiance* ($200 \rightarrow 400 \rightarrow 600 \rightarrow 800 \rightarrow 1000 \text{ W/m}^2$ step sequence), *step-change irradiance* ($1000 \rightarrow 600 \rightarrow 1000 \text{ W/m}^2$), *rapid double step* ($1000 \rightarrow 200 \rightarrow 1000 \text{ W/m}^2$, a stress test of extreme irradiance drop), *temperature step* ($25 \rightarrow 50 \rightarrow 25^\circ\text{C}$

at constant irradiance), *rapid fluctuation / cloud passage* (a continuous, piecewise-linear irradiance ramp), *sensor noise robustness* ($\pm 1\%$ Gaussian noise on the V, I measurements passed to `step()`, while the physical power recorded for metrics remains noise-free – noise models a flawed sensor, not a different physical plant), *partial shading (simple)* (two modules in series at $1000/400 \text{ W/m}^2$, producing two local maxima), and three *complex partial shading* patterns using three modules in series: Pattern A ($1000, 800, 600 \text{ W/m}^2$), Pattern B ($1000, 500, 300 \text{ W/m}^2$, widely spaced), and Pattern C ($1000, 1000, 400 \text{ W/m}^2$, the mildest of the four shading configurations, with only two effective local maxima).

B. Monte Carlo Methodology

Each algorithm-scenario pair is run for 50 independent Monte Carlo trials. Run k always uses random seed k regardless of which algorithm is being evaluated, so every algorithm sees an identical sequence of randomized initial duty cycles (and, where applicable, identical sensor-noise draws) at each run index – a paired design that both matches the intuitive requirement of a fair comparison and licenses the paired t -tests used in Section VII, rather than treating each algorithm's 50 runs as independent unpaired samples.

C. Q-Learning Train/Held-Out Tagging

Because Q-learning (Section V-D) is trained only on a single flat-irradiance module at three irradiance levels and STC temperature, every scenario is tagged *train* or *held-out* relative to that regime: a scenario counts as *train* only if every condition it exercises – irradiance level, temperature, absence of sensor noise, and single-module topology – matches what training actually used. By this criterion, only *steady-state* and *step-change irradiance* (both endpoints individually within the training irradiance set) qualify as train-distribution; the other nine scenarios are held-out, including all four partial-shading configurations, since Q-learning was never trained on a multi-module string at all regardless of whether the per-module irradiance values happen to lie within the training set. This is a deliberately coarse, condition-level classification: e.g. *step-change irradiance* is tagged *train* even though the mid-run irradiance *step itself* was never trained on, since training only ever sees static irradiance within an episode. Results for Q-learning are reported separately by this tag throughout Section VII and never pooled, directly addressing whether its policy generalizes or merely memorized training conditions.

D. Metrics

For every run, we compute: *tracking efficiency* $\eta = \int P_{\text{actual}} dt / \int P_{\text{theoretical max}} dt \times 100\%$; *convergence time*, the first time tracked power comes within 2% of the theoretical maximum (undefined, i.e. missing, if a run never converges within that tolerance); *settling time*, the first time tracked power *remains* within 2% continuously for at least 100 ms; *steady-state oscillation* (peak-to-peak and standard deviation of power after settling); *energy yield* and *energy yield ratio* (actual versus theoretical-maximum energy extracted); and

computational burden (mean wall-clock time per `step()` call, normalized to P&O). The 2% convergence threshold is a common convention in the MPPT literature. Non-convergence is common enough in this paper's partial-shading results (Section VII) that it is reported as an explicit *convergence rate* (fraction of runs converging) rather than silently averaged away – a run that never converges contributes no value to a convergence-time mean, and a naive mean over only the converged runs would misrepresent an algorithm that converges in, say, 2 of 50 runs as if it reliably converges quickly.

VII. RESULTS AND DISCUSSION

All results below are drawn directly from `results/summary_table.csv`, `results/statistical_tests.csv`, and `results/fuzzy_sensitivity_table.csv`, produced by the 50-run Monte Carlo sweep described in Section VI (19,260 raw metric rows for the five core algorithms plus a further 7,700 rows for the fuzzy rule-base sensitivity variants).

A. Steady-State Performance

Table III summarizes tracking efficiency, steady-state oscillation, and energy yield ratio at STC (1000 W/m^2 , 25°C). A one-way ANOVA across the five algorithms is significant for both tracking efficiency ($F = 11.65$, $p = 1.12 \times 10^{-8}$) and steady-state oscillation ($F = 21.88$, $p = 1.87 \times 10^{-15}$). Paired t -tests (paired by run, per Section VI's identical-seed design) confirm SMC's tracking efficiency is significantly higher than every other algorithm's (vs. P&O: $t = -5.11$, $p = 5 \times 10^{-6}$; vs. fuzzy logic: $t = -4.09$, $p = 1.6 \times 10^{-4}$; vs. IncCond: $t = -5.05$, $p = 6 \times 10^{-6}$; vs. Q-learning: $t = -3.63$, $p = 6.8 \times 10^{-4}$), and also has the lowest steady-state oscillation of all five.

TABLE III
STEADY-STATE PERFORMANCE (MEAN $\pm$ STD, $n = 50$)

Algorithm	Tracking Eff. (%)	Oscillation p2p (W)	Energy Ratio
P&O	99.73 ± 0.36	3.38 ± 1.01	0.9973 ± 0.0036
IncCond	97.64 ± 3.29	4.43 ± 1.18	0.9764 ± 0.0329
Fuzzy Logic	99.87 ± 0.18	3.14 ± 1.11	0.9987 ± 0.0018
Q-learning ^a	91.84 ± 15.70	2.88 ± 1.75	0.9184 ± 0.1570
SMC	99.90 ± 0.12	2.07 ± 1.29	0.9990 ± 0.0012

^aTraining-distribution scenario for Q-learning (see VI-C).

Q-learning's own training-distribution performance at steady state is notably more variable than the other four algorithms ($\sigma = 15.70$ versus ≤ 3.29 for every other algorithm) – this is not simply a held-out generalization gap, since steady-state is itself a training-distribution scenario. Its *convergence rate* at steady state is 80% (40 of 50 runs), versus 100% for P&O, fuzzy logic, and SMC and 70% for IncCond (Table IV); this should be reported alongside the held-out numbers below rather than only the generalization gap, since it shows the tabular policy does not fully solve even its own training scenario.

Fig. 1 shows tracking efficiency across all eleven scenarios; the two-line caption reflects that Q-learning's bars represent

held-out (untrained) conditions except at steady-state and step-change-irradiance, per Section VI-C's tagging.

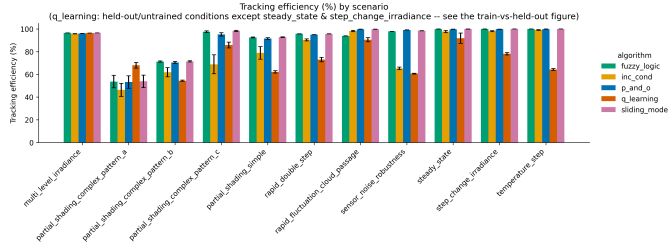


Fig. 1. Tracking efficiency (%) by scenario, all five core algorithms. Q-learning's bars reflect held-out (untrained) conditions except at *steady_state* and *step_change_irradiance*; see Fig. 4 for its dedicated train-vs-held-out comparison.

B. Partial Shading: A Shared Failure Across All Four Paradigms

Table IV reports *convergence rate* – the fraction of the 50 Monte Carlo runs reaching within 2% of the true global MPP – for all five core algorithms across all eleven scenarios. The result is unambiguous: every algorithm, across all four paradigms, achieves **0% convergence** on three of the four partial-shading configurations (*simple*, *Pattern A*, *Pattern B*). Only *Pattern C* – the mildest configuration, with only two effective local maxima – shows partial success, and even there IncCond converges in only 44% of runs. Fig. 2 shows this pattern across every scenario, not only the partial-shading ones.

TABLE IV
CONVERGENCE RATE (% OF 50 RUNS WITHIN 2% OF TRUE MPP)

Scenario	P&O	IncCond	Fuzzy	Q-learn.	SMC
Steady state	100	70	100	80 ^a	100
Multi-level irradiance	100	100	100	100	100
Step-change irradiance	100	100	100	100 ^a	100
Rapid double step	100	100	100	80	100
Temperature step	100	100	100	80	100
Rapid fluctuation	100	100	100	100	100
Sensor noise	100	70	100	70	100
Shading (simple)	0	0	0	0	0
Shading Pattern A	0	0	0	0	0
Shading Pattern B	0	0	0	0	0
Shading Pattern C	100	44	100	98	100

^aTraining-distribution scenario for Q-learning.

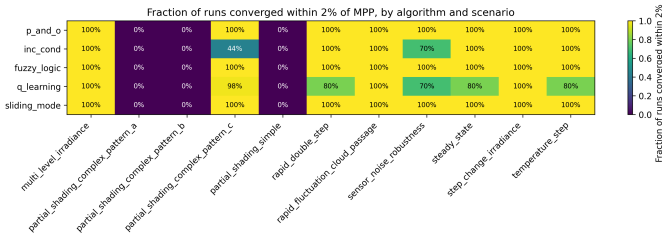


Fig. 2. Convergence rate (fraction of 50 runs reaching within 2% of the true MPP), algorithm $\times$ scenario. Colormap is *viridis* (perceptually uniform, colorblind-safe), deliberately not a red-green diverging map, which is exactly the wrong choice for the most common form of color blindness.

Multi-level irradiance originally showed a second, unrelated 0%-convergence result for IncCond and SMC specifically, discovered during this paper's own peer-review process and fixed prior to the results reported here – worth documenting because the mechanism is instructive. At the scenario's lowest level (200 W/m²), the true MPP ($V = 29.4$ V, $P = 46.7$ W) requires an effective source resistance $R_{in} = V/I = 18.6 \Omega$ – but a boost converter can only present $R_{in} = R_{load}(1-D)^2 \leq R_{load} = 9.2 \Omega$ (Section IV), so this MPP is unreachable at *any* valid duty cycle, not merely one outside this paper's $D \in [0.05, 0.90]$ clamp range. Every algorithm's duty cycle is driven toward the clamp floor while attempting to reach it. P&O tolerates this because it has no "hold" state at all – every step perturbs, so it keeps dithering right at the boundary until a perturbation away from it happens to increase power. IncCond and SMC, by contrast, both use "no voltage change between consecutive samples" as a proxy signal for being at or near the MPP (IncCond's $dV \approx 0$ boundary case; SMC's sliding surface $s = dP/dV$, which reads exactly zero when $dV = 0$) – and being clamped at the duty floor *also* produces $dV \approx 0$, for an entirely different reason (the duty cycle simply cannot move, not because the MPP condition is satisfied). Both algorithms' original implementations misread this as convergence, locked in a zero correction, and never attempted to move again even after irradiance later rose enough that the true MPP became reachable. Since the only way to tell "clamped" apart from "converged" is to actually try moving, both implementations now override a hold (or a correction that would push further into a limit already reached) with a small probe step away from that limit whenever duty is already sitting at a bound – using each algorithm's own existing step/correction scale, so behavior away from a boundary, where "hold" genuinely does mean converged, is unaffected. Table IV reports the result after this fix: both algorithms now reach 100% convergence on this scenario, matching P&O, fuzzy logic, and Q-learning, confirmed by direct trajectory re-inspection showing duty cycle actually escaping the clamp at each irradiance transition rather than staying frozen. See Section VIII for what this implies about the other results in this paper.

Crucially, the *mechanism* of failure differs by paradigm, and this is the more informative result than the raw 0% figure alone. Fig. 3 illustrates this for the *partial-shading-simple* scenario (two modules at 1000/400 W/m²; true global MPP at $V = 60.5$ V, $P = 206.4$ W). P&O, IncCond, fuzzy logic, and SMC all converge to the *same local maximum* (≈ 193 W, $\sim 6.5\%$ below the true global optimum) – unsurprising, since all four follow the local $dP/dV = 0$ (or equivalent) condition and have no mechanism to escape a local maximum once reached. Q-learning, which was never trained on any multi-module configuration at all (Section VI-C), does something qualitatively *worse*: rather than converging to either maximum, it moves the operating point to a low-power region (≈ 114 W, roughly 45% below even the local maximum the other four algorithms reach) and oscillates there indefinitely. This is a materially different – and more severe – failure mode than "stuck at a local maximum." Reporting Q-learning's held-out performance honestly, rather than only its training-distribution

numbers, is precisely what surfaces this: a real, qualitatively distinct weakness rather than a merely quantitative one.

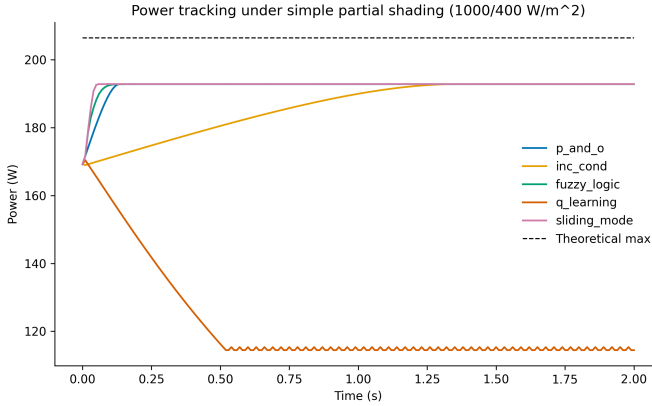


Fig. 3. Power tracking trajectories under simple partial shading ($1000/400 \text{ W/m}^2$). P&O, fuzzy logic, and SMC converge to a shared local maximum ($\approx 193 \text{ W}$); IncCond converges more slowly to the same point; Q-learning (untrained on this topology) moves to and oscillates around a lower-power operating point ($\approx 114 \text{ W}$) instead.

C. Q-Learning: Train-Distribution vs. Held-Out

Fig. 4 compares Q-learning’s tracking efficiency across all scenarios it was trained on versus all scenarios it was not, per the tagging methodology in Section VI-C. Consistent with Section VII-B, the largest held-out gaps occur under partial shading (a topology never seen in training, not merely an unseen irradiance level), underscoring that this paper’s train/held-out split – unlike flat-irradiance-only held-out evaluation – is specifically designed to expose a generalization failure that a same-topology-only evaluation would miss entirely.

Notably, *temperature_step* – a single-dimension change (temperature only, same single-module topology, same irradiance) rather than a topology change – produces a held-out tracking efficiency of only 64.3%, comparable in severity to two of the four partial-shading patterns and well below every other held-out scenario except *sensor_noise_robustness* and *partial_shading_simple*. This is not merely an unseen-condition penalty: convergence rate for this scenario is unchanged from before the temperature-step-away portion of the run (80%, identical to the other held-out scenarios’ pre-step behavior), but tracking efficiency collapses because the agent enters sustained, large-amplitude oscillation after the $25 \rightarrow 50^\circ\text{C}$ step (steady-state oscillation p2p rises from a few watts to $\sim 100 \text{ W}$) – consistent with its fixed voltage-bin discretization (Section V-D), calibrated only against STC-temperature operating ranges, placing the post-step operating point in a state region the training distribution never adequately explored. This strengthens, rather than merely restates, this paper’s central generalization finding: even a single-dimension shift outside the training distribution – not only a full topology change – is enough to produce a severe, qualitatively distinct failure mode for the tabular Q-learning design used here.

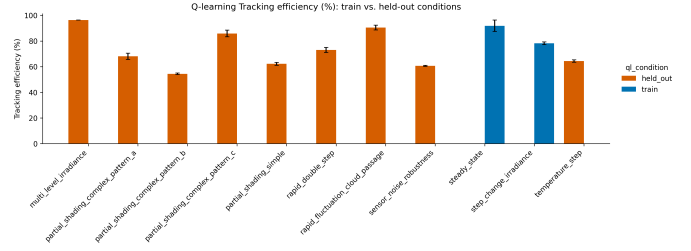


Fig. 4. Q-learning tracking efficiency (%), grouped by whether the scenario’s conditions matched its training distribution.

D. Fuzzy Rule-Base Sensitivity

Table V reports steady-state tracking efficiency for the full 7×7 rule base against reduced 5×5 and 3×3 variants (Section V). The 5×5 variant’s 95% confidence interval ($[99.83, 99.93]$) overlaps almost entirely with the full 7×7 base’s ($[99.82, 99.92]$), i.e. the two are statistically indistinguishable at steady state – real support for the “reduced rule base matches full performance” contribution this sensitivity analysis was designed to test. The 3×3 variant is measurably, though modestly, lower (99.01%, CI $[98.58, 99.43]$). This result has direct practical relevance given the computational-burden finding below: fuzzy logic’s 49-rule Mamdani inference is measured at roughly $101\times$ P&O’s per-step cost (Table VI), so a rule-base reduction that preserves performance directly reduces that overhead. Fig. 5 shows the full per-scenario comparison, not only the steady-state figures quoted above.

TABLE V
FUZZY RULE-BASE SENSITIVITY: STEADY-STATE TRACKING EFFICIENCY (%)

Rule Base	Mean	95% CI
3×3	99.01	[98.58, 99.43]
5×5	99.88	[99.83, 99.93]
7×7 (full)	99.87	[99.82, 99.92]

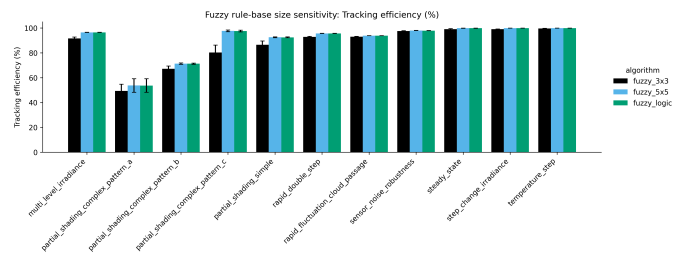


Fig. 5. Fuzzy rule-base size sensitivity: tracking efficiency (%) across all eleven scenarios, 3×3 vs. 5×5 vs. full 7×7 .

E. Computational Burden

Table VI reports mean $\text{per_step}()$ wall-clock time, normalized to P&O. Fuzzy logic’s Mamdani inference and centroid defuzzification over a 201-point grid make it roughly two orders of magnitude more expensive than P&O; Q-learning’s table lookup is substantially cheaper than fuzzy logic but still

an order of magnitude above P&O, reflecting per-step NumPy discretization overhead rather than the lookup itself. IncCond remains close to P&O's baseline cost; SMC is somewhat above it, the extra cost of the boundary-clamp check added in Section VII-B, as Fig. 6 shows directly. Wall-clock time is a practical, deployment-relevant metric, but it conflates algorithmic complexity with implementation detail (NumPy vectorization overhead, Python interpreter cost, and run-to-run timing variance – SMC's measurement here is more sensitive to that variance than the others', since its per-step cost is small in absolute terms) that an operation-count or FLOP-based metric would isolate; we report wall-clock time here because it is what a real embedded MPPT controller's sample-period budget actually constrains, but treat the relative orders-of-magnitude (not the exact multipliers) as the robust takeaway.

TABLE VI
COMPUTATIONAL BURDEN (RELATIVE TO P&O = 1.0)

Algorithm	Relative Burden
P&O	1.00
IncCond	0.77
SMC	1.61
Q-learning	10.86
Fuzzy Logic	101.26

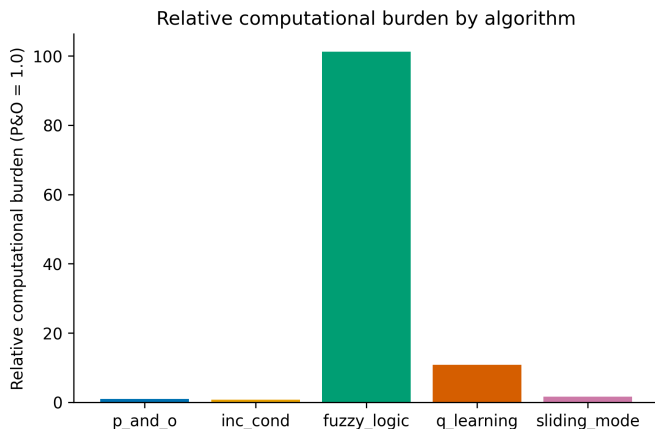


Fig. 6. Relative computational burden by algorithm (P&O = 1.0).

F. Summary

No single algorithm dominates across every scenario and metric: SMC gives the best steady-state tracking efficiency and lowest oscillation at a computational cost still well below the fuzzy and learning-based paradigms, but it (and every other algorithm, regardless of paradigm) fails to find the true global MPP under all but the mildest partial-shading pattern tested. Fuzzy logic's rule-reduction result is a genuine, statistically-supported contribution, but its absolute computational cost remains two orders of magnitude above the perturbative baselines even after reduction. Q-learning's generalization failure under partial shading is qualitatively worse than the other paradigms' local-maximum failures, a direct consequence of its training distribution never including multi-module shading

– and the same fixed discretization is fragile even to a single-dimension held-out condition (temperature), not only a topology change, as Section VII's train-vs-held-out comparison shows. See Section VIII for the limitations this implies.

VIII. CONCLUSION

This paper compared five MPPT algorithms spanning four genuinely distinct design paradigms – perturbative, rule-based, learning-based, and model-based nonlinear control – under a common, open-source, statistically rigorous simulation framework, validated against a real commercial PV datasheet. Sliding mode control achieved the best steady-state tracking efficiency and lowest oscillation at a computational cost still an order of magnitude below the fuzzy and learning-based paradigms. A reduced 5×5 fuzzy rule base statistically matched the full 7×7 base's performance, a genuine rule-reduction contribution given fuzzy logic's substantial computational overhead. Most significantly, every algorithm across all four paradigms – including the model-based SMC controller with its Lyapunov stability guarantee – failed to reach the true global maximum power point in the majority of complex partial-shading patterns tested, with the failure mode differing meaningfully by paradigm: the perturbative and model-based algorithms converge to a shared local maximum, while Q-learning (never trained on multi-module shading) fails in a qualitatively worse way, moving to and oscillating around a substantially lower-power operating point instead. This paper's own peer review process also surfaced and fixed a genuine implementation bug in IncCond and SMC's boundary-case logic (Section VII-B), which had misread being clamped at a hard duty-cycle limit as convergence and never recovered once irradiance later made the true MPP reachable again – the corrected results reported throughout this paper reflect the fix, not the original defect.

A. Limitations

Several limitations should inform how these results are read and motivate future work. First, the boost converter enters discontinuous conduction mode (DCM) at light load (Section IV), where the small-signal model every algorithm's duty-cycle-perturbation reasoning implicitly relies on no longer holds; whether this contributes to any algorithm's degraded performance specifically in the low-irradiance and partial-shading scenarios has not been isolated from the algorithms' intrinsic global-search limitations, and would require a dedicated ablation to disentangle. Second, low-irradiance validation (Section III-D) is necessarily coarser than the NOCT point, since the datasheet gives only a single efficiency-retention claim at 200 W/m^2 rather than a full I-V table; the model's measured 93.5% retention is close to, but not an exact match for, the datasheet's claimed 95.5%, plausibly because this simplified model does not scale shunt resistance with irradiance as a full five-parameter model would. Third, Q-learning's tabular state discretization and training protocol (Section V-D) are one specific design point in a much larger learning-based MPPT design space; whether a different state representation, function approximation, or a training distribution that includes partial

shading would close the gap observed in Section VII-B is an open question this paper does not attempt to answer, since doing so changes the fair-comparison premise of evaluating the same tabular design used throughout. Fourth, this is a simulation-only study; while the two-diode model is quantitatively validated against a real datasheet (Section III-D), hardware validation would strengthen any of these conclusions further.

B. Future Work

The clearest direction suggested by this paper's own results is a deeper, dedicated study of learning-based MPPT under partial shading – explicitly training on shading patterns rather than only flat irradiance, and comparing tabular Q-learning against function-approximation and deep-RL alternatives such as recent fuzzy-augmented actor-critic designs [29] – since the generalization gap found here is a training-data problem, not necessarily a fundamental limitation of learning-based control. A second, complementary direction is a fair, reproducible benchmark of metaheuristic global-search MPPT methods (which this paper deliberately excludes, to avoid the saturated "yet another bio-inspired optimizer" pattern) against this paper's partial-shading scenarios directly, since metaheuristics are specifically designed for the global-search problem this paper's results show every non-metaheuristic paradigm struggles with. Finally, the DCM/light-load confound flagged above is a concrete, answerable question – rerunning the low-irradiance and partial-shading scenarios with a converter design that stays in CCM across the full operating range would isolate whether DCM sensitivity contributes to any algorithm's measured performance independent of its global-search capability.

Code and Data Availability

All simulation code, raw Monte Carlo results, and figures are released under the MIT License at <https://github.com/Deugs/mppt-comparison> and archived with a citable DOI at [Zenodo DOI -- fill in once minted].

REFERENCES

- [1] T. Esmar and P. L. Chapman, "Comparison of photovoltaic array maximum power point tracking techniques," *IEEE Transactions on Energy Conversion*, vol. 22, no. 2, pp. 439–449, 2007.
- [2] N. Femia, G. Petrone, G. Spagnuolo, and M. Vitelli, "Optimization of perturb and observe maximum power point tracking method," *IEEE Transactions on Power Electronics*, vol. 20, no. 4, pp. 963–973, 2005.
- [3] K. H. Hussein, "Maximum photovoltaic power tracking: An algorithm for rapidly changing atmospheric conditions," *IEE Proceedings - Generation, Transmission and Distribution*, vol. 142, no. 1, pp. 59–64, 1995.
- [4] M. H. Ibrahim, S. P. Ang, M. N. Dani, M. I. Rahman, R. Petra, and S. M. Sulthan, "Optimizing step-size of perturb & observe and incremental conductance mppt techniques using pso for grid-tied pv system," *IEEE Access*, vol. 11, pp. 13 079–13 090, 2023.
- [5] T. Zhu, J. Dong, X. Li, and S. Ding, "A comprehensive study on maximum power point tracking techniques based on fuzzy logic control for solar photovoltaic systems," *Frontiers in Energy Research*, vol. 9, p. 727949, 2021.
- [6] C. B. Nzoundja Fapi, H. Tchakounté, M. Ndje, P. Wira, D. Ould Abdelham, M. Louzani, and M. Kamta, "Fuzzy logic-based maximum power point tracking control for photovoltaic systems: A review and experimental applications," *Archives of Computational Methods in Engineering*, vol. 32, no. 4, pp. 2405–2428, 2025.
- [7] S. K. Sunkara and K. Balakrishna, "A novel design and analysis of hybrid fuzzy logic mppt controller for solar pv system under partial shading conditions," *Scientific Reports*, vol. 14, no. 1, p. 10256, 2024.
- [8] P. Kofinas, S. Doltsinis, A. I. Dounis, and G. A. Vouros, "A reinforcement learning approach for mppt control method of photovoltaic sources," *Renewable Energy*, vol. 108, pp. 461–473, 2017.
- [9] Y. Chen, D. Lin, F. Xu, X. Li, W. Wang, and S. Ding, "Research on q-table design for maximum power point tracking-based reinforcement learning in pv systems," *Energies*, vol. 16, no. 21, p. 7286, 2023.
- [10] B. C. Phan, Y.-C. Lai, and C. E. Lin, "A deep reinforcement learning-based mppt control for pv systems under partial shading condition," *Sensors*, vol. 20, no. 11, p. 3039, 2020.
- [11] V. I. Utkin, "Variable structure systems with sliding modes," *IEEE Transactions on Automatic Control*, vol. 22, no. 2, pp. 212–222, 1977.
- [12] F. F. Ahmad, C. Ghenai, A. K. Hamid, and M. Bettayeb, "Application of sliding mode control for maximum power point tracking of solar photovoltaic systems: A comprehensive review," *Annual Reviews in Control*, vol. 49, pp. 173–196, 2020.
- [13] W. Gao and J. C. Hung, "Variable structure control of nonlinear systems: A new approach," *IEEE Transactions on Industrial Electronics*, vol. 40, no. 1, pp. 45–55, 1993.
- [14] J.-J. E. Slotine and W. Li, *Applied Nonlinear Control*. Englewood Cliffs, NJ: Prentice Hall, 1991.
- [15] L. Liu, "Photovoltaic mppt control and improvement strategies considering environmental factors: Based on pid-type sliding mode control and improved grey wolf optimization," *Measurement and Control*, vol. 58, no. 2, 2025.
- [16] R. W. Erickson and D. Maksimovic, *Fundamentals of Power Electronics*, 2nd ed. Springer, 2001.
- [17] F. M. Serra, F. D. Esteban, and O. D. Montoya, "Control of dc-dc boost converter in discontinuous conduction mode feeding a constant power load," *Results in Engineering*, vol. 23, p. 102732, 2024.
- [18] M. G. Villalva, J. R. Gazoli, and E. Ruppert Filho, "Comprehensive approach to modeling and simulation of photovoltaic arrays," *IEEE Transactions on Power Electronics*, vol. 24, no. 5, pp. 1198–1208, 2009.
- [19] K. Ishaque, Z. Salam, and H. Taheri, "Simple, fast and accurate two-diode model for photovoltaic modules," *Solar Energy Materials and Solar Cells*, vol. 95, no. 2, pp. 586–594, 2011.
- [20] A. M. Syed, S. K. Matam, C. Yammani, and S. R. Sathigari, "Parameter extraction of two diode photovoltaic model using an analytical approach," *IEEE Latin America Transactions*, vol. 22, no. 5, 2024.
- [21] E. Moshksar, "Two datasheet-based approaches for double-diode pv modeling with an efficient trade-off between accuracy and simplicity," *Heliyon*, vol. 10, no. 16, p. e36424, 2024.
- [22] L. Abad, S. Tamalouzt, K. Djermouni, S. Mekhilef, and Y. Belkhier, "Analytical modeling of photovoltaic systems under partial shading conditions incorporating bypass and blocking diodes influence," *International Journal of Energy Research*, vol. 2025, no. 1, 2025.
- [23] M. S. Endiz, G. Gökküş, A. E. Coşgun, and H. Demir, "A review of traditional and advanced mppt approaches for pv systems under uniformly insolation and partially shaded conditions," *Applied Sciences*, vol. 15, no. 3, p. 1031, 2025.
- [24] S. J. Yaqoob, H. Arnoos, N. T. Alwan, M. Bajaj, I. M. Alwan, M. H. A. Aljanabi, B. A. Zneid, and M. S. Gernew, "Advanced maximum power point tracking in photovoltaic systems: A comprehensive review of classical, ai-based, and metaheuristic optimization techniques," *Engineering Reports*, vol. 7, no. 9, 2025.
- [25] F. Abdelmalek, H. Afghoul, F. Krim, D. E. Zabia, H. Trabelsi, M. Bajaj, I. Zaitsev, and V. Blazek, "Experimental validation of effective zebra optimization algorithm-based mppt under partial shading conditions in photovoltaic systems," *Scientific Reports*, vol. 14, no. 1, 2024.
- [26] S. A. Taha, Z. S. Al-Sagar, M. A. Abdulsada, M. Alruwaili, and M. A. Ibrahim, "Design of an efficient mppt topology based on a grey wolf optimizer-particle swarm optimization (gwo-pso) algorithm for a grid-tied solar inverter under variable rapid-change irradiance," *Energies*, vol. 18, no. 8, p. 1997, 2025.
- [27] S. M. Ashraf, M. S. B. Arif, M. Khouj, S. M. Ayob, and M. I. Masud, "Python-based implementation of metaheuristic mppt techniques: A cost-effective framework for solar photovoltaic systems in developing nations," *Energies*, vol. 18, no. 12, p. 3160, 2025.
- [28] W. De Soto, S. A. Klein, and W. A. Beckman, "Improvement and validation of a model for photovoltaic array performance," *Solar Energy*, vol. 80, no. 1, pp. 78–88, 2006.
- [29] D. Ortiz-Munoz, D. Luviano-Cruz, L. A. Perez-Dominguez, A. G. Rodriguez-Ramirez, and F. Garcia-Luna, "Hybrid fuzzy-ddpg approach for efficient mppt in partially shaded photovoltaic panels," *Applied Sciences*, vol. 15, no. 9, p. 4869, 2025.